ARNet: Prior Knowledge Reasoning Network for Aircraft Detection in Remote-Sensing Images

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***Abstract*— Amidst the landscape of contemporary remote-sensing (RS) technology, the endeavor to detect and recognize aircraft within RS images (RSIs) assumes pivotal strategic and practical significance. The complex nature of fine-grained aircraft recognition is a result of the intricate interplay between aircraft and their background environments, alongside category imbalance, which collectively leads to the emergence of a long-tail distribution within the dataset. However, experts proficient in RSI interpretation can effectively address these challenges through the application of prior knowledge. This article introduces the aircraft reasoning network (ARNet), a framework tailored for aircraft detection and fine-grained recognition in RSIs, building upon prior knowledge employed in expert interpretation. Specifically, the knowledge reasoning module (KRM) introduces a knowledge graph (KG) that incorporates both common and expert knowledge into the end-to-end network. Additionally, the network encompasses a spatial context module (SCM) and an airport facility relationship module (AFRM). These components facilitate highly accurate detection and recognition of fine-grained aircraft in diverse environmental contexts by employing adaptive prior knowledge reasoning and optimizing target spatial location. Furthermore, an independent aircraft component discrimination module (ACDM) distinguishes aircraft based on their predominant component features, contributing to improved classification performance in both the few-shot and easily confused categories. Moreover, this article introduces the AR-RSI dataset, a compilation of RSIs capturing fine-grained aircraft targets from diverse locations. The effectiveness and superiority of ARNet are exemplified by AR-RSI, achieving a minimum of 3.7% higher mean average precision (mAP) than the mainstream aircraft detection framework.**

***Index Terms*— Aircraft component, aircraft detection and fine-grained recognition, prior knowledge in knowledge graph (KG), remote-sensing images (RSIs).**

I. Introduction

RAPID advancements in remote-sensing (RS) technology have enabled modern airborne and satellite sensors to efficiently capture RS images (RSIs) with improved resolution and enhanced clarity. Artificial intelligence (AI) is assuming an increasingly pivotal role in the field of RS, drawing significant interest from researchers and resulting in a series of critical advancements. Extensive research has been conducted to meet the demand for automated analysis of data, with a particular focus on deep-learning (DL)-based object detection algorithms. Among these algorithms, the detection of aircraft targets holds a vital position within the aviation and detection domains. However, large-scale RSIs often exhibit complex and high-density environmental backgrounds. Additionally, the imbalanced distribution of fine-grained aircraft categories presents pervasive challenges related to long-tailed distributions and small-sample problems. Object detection inherently demands precision and robustness, making aircraft detection in RSIs a persistent challenge.

Traditional aircraft detection methods rely heavily on manually designed complex feature representation techniques or the challenging tasks of parameter tuning and function debugging, including algorithms such as contour detection [1], template matching [2], scale-invariant feature transform (SIFT) features [3], invariant local features [4], histogram of oriented gradient

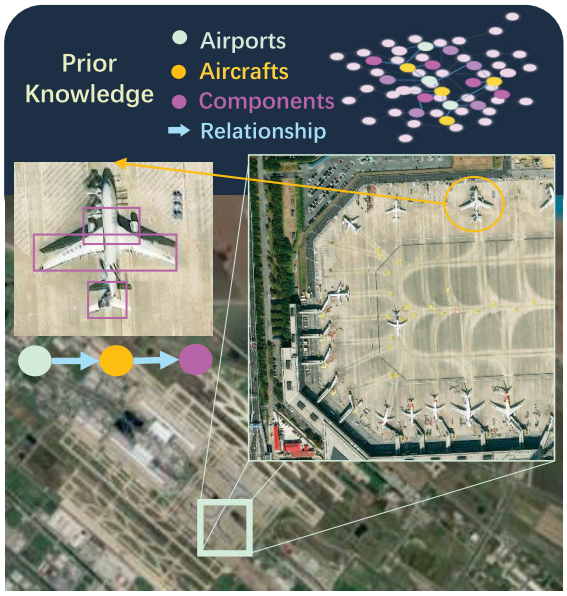


Fig. 1. Illustration of how the proposed prior knowledge reasoning network enhances the detection and recognition of aircraft in RSIs. The targeted reasoning process is achieved by efficiently describing a KG that contains common and expert knowledge, such as relationships and characteristics between airports, aircraft, and their components. This knowledge is then seamlessly integrated into the detection and recognition pipeline.

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(HOG) feature pyramids [5], and support vector machines (SVMs) [6]. Extracting generic features becomes highly challenging in scenarios characterized by complex backgrounds and diverse target categories.

The growing research interest in AI and DL has led to the widespread application of object detection methods based on convolutional neural networks (CNNs) in RSIs. The horizontal bounding box (HBB) [7] and oriented bounding box (OBB) [8] are common detectors employed in CNN-based aircraft detection algorithms. Several recent studies have developed algorithms and models to improve detection performance by utilizing the characteristics of targets in RSIs. For instance, Sun et al. [9] employed various algorithms, including multimodal image fusion and joint learning, while Wang et al. [10] integrated clustering features of RS targets into the proposed target region proposals of the detection network. Liu et al. [11] introduced an aircraft detection scheme that combines angle clustering with CNNs, using mean shift clustering to generate candidate target regions based on detected corners in binary images. However, despite efforts to enhance aircraft detection in RSIs by increasing target diversity and improving local features, these algorithms struggle with stability and robustness in diverse environments and imbalanced data, often relying on manual interventions for data selection and feature clustering, which can compromise their overall performance.

Incorporating human prior knowledge or experience into the network offers an effective approach to minimize manual intervention and enhance target feature extraction. For example, Hu et al. [12] introduced an end-to-end target detection algorithm that effectively captures the interrelationships between appearance features and object geometry. To improve algorithmic accuracy, Li et al. [13] employed multispectral and hyperspectral image fusion from different modalities. Chen et al. [14] and Xu et al. [15] utilized human-guided networks grounded in common sense for large-scale detection tasks, achieving promising outcomes on the COCO dataset. However, most contemporary algorithms for aircraft detection and recognition lack knowledge-driven reasoning. Inspired by the application of a common sense knowledge graph (KG) to enhance target learning, as demonstrated in [15], our research establishes a framework for fine-grained aircraft detection and recognition in RSIs. This article addresses the challenge of interpreting aircraft, focusing on three key aspects.

1) *Imbalanced Data Due to Long-Tail Distribution and Limited Samples:* In RSIs, the detection and recognition of aircraft often confront challenges related to imbalanced training data and a scarcity of samples for specific aircraft categories. For example, at the majority of domestic airports, there is a notable surplus of short- and medium-haul passenger aircraft in comparison to long-haul passenger aircraft. This imbalance becomes even more pronounced when considering the relatively low presence of early warning aircraft with airborne early warning and control systems (AWACS) and refueling aircraft.
2) *Fine-Grained Aircraft Recognition:* In contrast to standard target detection datasets like COCO, the recognition of aircraft targets necessitates a fine-grained approach. This is due to the subtle interclass feature distinctions among targets. Aircraft categories, such as transport aircraft, refueling aircraft, and AWACS, often undergo modifications that make them prone to confusion with prototype aircraft. Moreover, there are distinctions within intraclass features. Aircraft targets within RSIs exhibit significant variations in size and appearance from various angles and viewpoints, resulting in a diversity of visual characteristics within the same category.
3) *Target Density and Diversity of RS Scenes:* In large-scale RSIs, aircraft targets can be found in both dense clusters and relatively sparse arrangements. Aircraft may be situated among runways, lush vegetation, architectural structures, and various other features. This wide range of density and diversity in RS scenes significantly increases the challenges associated with aircraft target detection.

To tackle the challenges associated with aircraft detection and recognition, this study presents an aircraft reasoning network (ARNet), a knowledge-based reasoning framework as illustrated in Fig. 1 created for the detection and recognition of aircraft in RSIs. The contributions of this article can be summarized as follows.

1) *Knowledge-Based Reasoning Network Architecture:* In addressing the challenge of fine-grained aircraft detection in RSIs, particularly in the presence of long-tailed data issues, this study introduces the knowledge reasoning module (KRM). Its primary aim is to improve the accuracy of fine-grained aircraft recognition by establishing semantic relational connections between small and frequently occurring samples. This is accomplished through the use of a KG that incorporates both common and expert knowledge.
2) *Relationship-Based Location Optimization:* To tackle the challenges presented by varying target densities and diverse scenarios, this study introduces the spatial context module (SCM) and the airport facility relationship module (AFRM). These modules are designed to leverage the positional alignment among different aircraft categories and to capture the spatial contextual relationships between aircraft and airport facilities.
3) *Independent Aircraft Component Discrimination:* To enhance fine-grained aircraft classification, particularly in the few-shot categories, and to manage the complexities of easily confusable categories, the study utilizes the aircraft component discrimination module (ACDM). ACDM leverages dominant component features of aircraft to improve classification performance.

The structure of the subsequent sections of this article will be as follows. Section II provides insights into related work, Section III delves into technical details, Section IV introduces the dataset and presents experimental results, and finally, Section V offers a summary of this article.

II. Related Works

A. Aircraft Detection and Recognition in RSIs

Researchers have recently focused their efforts on achieving rapid and precise localization and recognition of aircraft within RSIs through DL methods. Wu et al. [16] utilized candidate target regions to detect aircraft in RSIs. Li et al. [17] achieved

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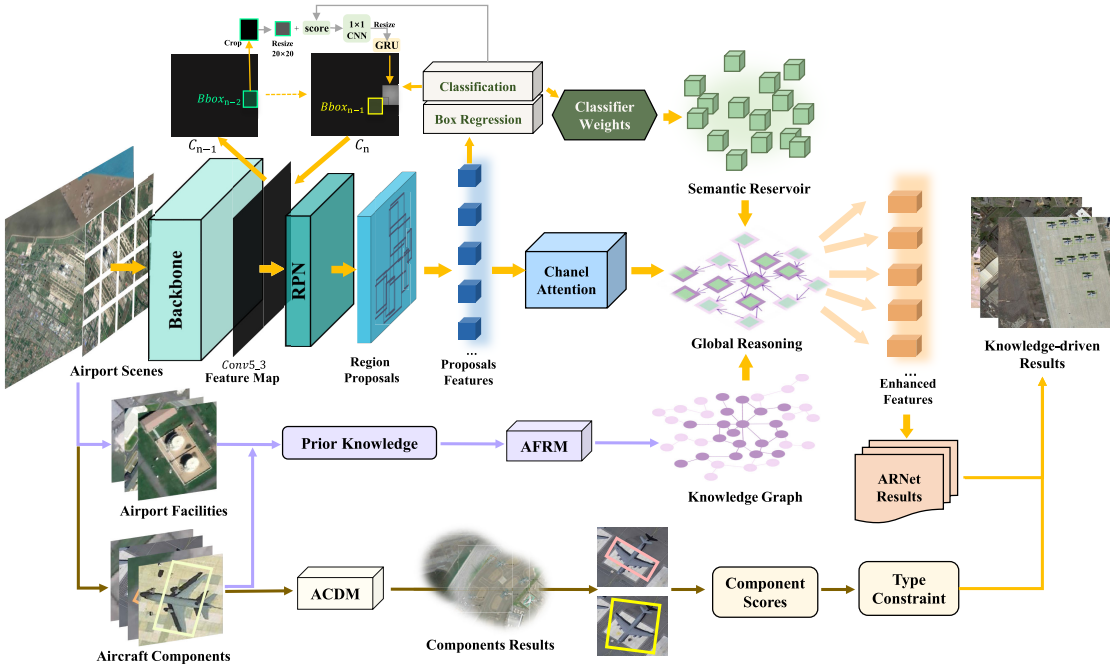


Fig. 2. ARNet framework for fine-grained aircraft reasoning is an end-to-end network designed for both detection and recognition.

aircraft detection through reinforcement learning approaches. Additionally, Zhao et al. [18] introduced a method that predicts key aircraft points and utilizes their coordinates to determine the aircraft's orientation, category, and location. The OE-FCN [19], a novel end-to-end network, facilitates aircraft detection by employing a feature mining mechanism to capture intraclass variance within RSIs. Zhang et al. [20] proposed a weakly supervised aircraft detection algorithm tailored for high-resolution, clear RSIs. The introduction of X-LineNet [21], a detection model translating aircraft detection in RSIs into intersecting line segments, enhances the accuracy of detection results. The components and structural attributes of an aircraft are crucial factors in the process of aircraft type recognition. Wang et al. [22] employed principal component analysis (PCA) to examine the structural and component attributes of aircraft targets, subsequently employing SVMs for the classification of processed targets. Jia et al. [23] designed an additional discriminative module specifically tailored to distinguish the distinctive fuselage rear section characteristics of the ARJ21 from other potentially confounding aircraft types. Zhao et al. [18], Liu et al. [24], Zuo et al. [25], and Wu et al. [26] identified a set of key landmarks or structural components to determine the outlines and positions of aircraft, utilizing CNNs or template matching for the classification task. Zhao et al. [18] employed eight specific landmarks to determine the structural characteristics and component positions of aircraft, while Zuo et al. [25] detected four key points on the nose, tail, and both wingtips and applied optimized contour masks. Additionally, Chen et al. [27] and Dong et al. [28] utilize the segmentation network algorithm to achieve the recognition task. Chen et al. [27] partitioned and annotated the rigid aircraft structure into masks corresponding to five distinct segments: the nose, left wing, right wing, fuselage, and tail. Subsequently, they applied addition operations and an attention mechanism, combining five features to determine the aircraft's type.

B. Visual Reasoning

Visual reasoning refers to the application of DL technology to the automatic extraction and reasoning of information such as objects, scenes, and relationships in images. The visual question answering (VQA) model [29] is one of the most common working paradigms for visual reasoning, where a textual answer is provided after understanding a given image. Target detection visual reasoning tasks typically use multimodal data, such as by using structured knowledge in [30] to guide CNNs for image reasoning. Image interpretation and information fusion [9] from different sensors or multiple medical image sources [31]. In addition, Cong et al. [32] and Tian et al. [33] utilize attention mechanisms and sequential relationships to achieve target detection reasoning tasks. The cornerstone of visual reasoning algorithms lies in leveraging external knowledge to guide the network in conducting inference. However, RS data exhibit distinct variations from optical and medical images in terms of target characteristics. Thus, an approach grounded in the distinctive attributes of RS data is essential to achieve precise target interpretation.

C. Fine-Grained Recognition

Current approaches to fine-grained classification primarily rely on visual information to address the fine-grained classification challenge. Alternatively, some methods attempt to leverage multimodal data to construct a unified representation for fine-grained recognition. ACDN [23] enhances classification performance by accurately identifying aircraft with

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dominant part features. MA-CNN [34] introduces a multiattention model that synergistically reinforces part generation and feature learning. NTSNet [35] presents a self-supervised mechanism that efficiently recognizes unlabeled regions. DBT-Net [36] introduces deep bilinear transformation, effectively utilizing semantic information for acquiring bilinear features. CAP [37] employs context-aware attentional pools to capture subtle image variations. Geo-Aware [38] integrates geographic information before fine-grained classification, using it for post-processing and feature modulation. KERL [39] combines rich additional information with a deep neural networks (DNNs) architecture to organize diverse visual concepts within a KG framework. CVL [40] introduces a two-branch network that separately learns visual and textual features, culminating in the fusion of these two facets to yield a final latent semantic representation.

III. Methodology

A. Overview

This section presents the ARNet framework, as illustrated in Fig. 2, for aircraft target detection and recognition. It emphasizes the integration of four essential modules: SCM captures spatial correlations among aircraft targets. KRM incorporates prior knowledge into the network using a KG, that combines expert insights and common knowledge, contributing to improved detection and recognition results. ACDM enhances fine-grained aircraft recognition, by leveraging dominant component features, particularly for few-shot and easily confusable categories. AFRM correlates fine-grained aircraft with airport facilities, ultimately improving target localization.

B. Spatial Context Module

The employment of the SCM module serves to infer spatial context at the instance level. The incorporation of spatial context enables the detection process to acquire information from both the feature map and the target context. The primary function of SCM is to maintain a record of previously detected target locations with a high degree of confidence, thus improving the detection of aircraft in nearby or associated regions. Furthermore, it can leverage the distinctive spatial patterns that aircraft adhere to within airport environments. This process reflects the human tendency to identify the most assured target in an airport first, followed by considerations of the arrangement pattern and differences between instances.

For an airport scene s_a, the objective function of training model $\mathcal{M}$ can be described as the maximum log-likelihood of the loss function

$$\begin{aligned}\arg\max_{\mathcal{M}} \mathcal{L} &= \log \mathbb{P}(T_{1:N} \mid \mathcal{M}, s_a) \\ &= \sum_{n=1:N} \log \mathbb{P}(T_n \mid \mathcal{T}_{0:n-1}, \mathcal{M}, s_a) \end{aligned} \tag{1}$$

where $\mathcal{T} = [T_1, T_2, \ldots, T_N]$ contains both bounding boxes and categories of detected instance targets in the airport scene s_a, and T_0 is the empty set. N is the maximum number of targets. Equation (1) is approximated by detecting each target separately in the region proposal-based detector. The spatial context C_{n-1} serves as a repository for retaining previously acquired targets. Whenever a new target is detected, C_n will be updated. C_{n-1} will be optimized and updated together with the detection model $\mathcal{M}$, (1) can be approximated as

$$\arg\max_{\mathcal{M},C} \mathcal{L} \approx \sum_{n=1:N} \log \mathbb{P}(T_n \mid C_{n-1}, \mathcal{M}, s_a). \tag{2}$$

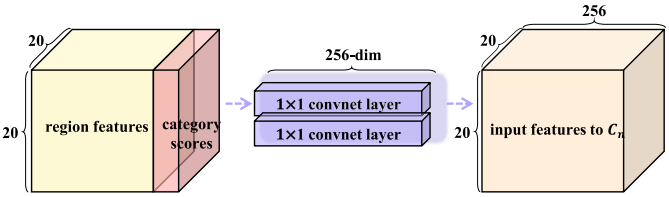


Fig. 3. Spatial context and classifier scores are incorporated into the target region within the pseudo-image. Two convolutional layers are employed to integrate these two component features.

Following the derivation of the aforementioned formulas, the establishment of a storage unit akin to a feature map becomes imperative to streamline the convolution operation. This storage unit, which preserves highly scored instances, emulates an image representing accurate detection outcomes and is accordingly denominated as a pseudo-image. It is worth noting that the context related to the aircraft target remains unaffected by the input image size, initially set at 20×20. The dimensions of C_n can be adapted via bilinear interpolation during the learning process. Subsequently, the pertinent content requiring storage within C_n is identified. As depicted in Fig. 3, cropping and resizing are employed to generate suitable feature dimensions, including contextual features that maintain spatial relationships among aircraft, as well as category scores, which are crucial for evaluating high-confidence targets. Two 1×1 convolutional layers are utilized to merge classification scores and spatial context features, resulting in the creation of the composite feature denoted as x_t.

C_n is then updated iteratively in subsequent stages of the learning process using the convolutional gated recursive unit (GRU). Within the GRU, there exists an update gate z_t and a reset gate r_t. The sigmoid function, $\sigma(\cdot)$, is employed to evaluate the influence of the previous hidden state, h_{t-1}, and the input data, x_t, in generating the new candidate hidden state, $\tilde{h}_t$, as well as determining the extent to which the current state, h_t, is preserved

$$z_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{3}$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \tag{4}$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h r_{t-1} \odot h_{t-1} + b_h) \tag{5}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \tag{6}$$

After the GRU operation, the revised spatial context is combined with the previous content and then reintegrated into C_n. During this update, a greedy strategy is employed to prioritize instance targets with high reliability as valuable foreground targets. It is important to note that this strategy might not align precisely with the actual reference context used by humans for distinguishing aircraft, especially during network learning. Nonetheless, the subsequent KRM includes other relational properties.

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C. Knowledge Reasoning Module

The KRM of ARNet demonstrates flexibility in assimilating diverse knowledge types through the KG, which is consistently shared during both the training and testing phases. The comprehensive description of the KRM unfolds in three key segments: First, the KG is translated into a trainable representation of the network. Subsequently, the target features are enhanced through knowledge-based reasoning. Lastly, an attention mechanism is deployed to reduce global noise and facilitate feature integration.

1) Specific Relation and Attribute KG: KG is employed to establish semantic connections among specific aircraft categories. These instances encompass various relationships, including "is-kind-of" (such as "Boeing737" and "civilian airliner"), "is-part-of" (such as "aerofoil" and "Boeing737"), and "similarity" (such as Boeing airliner and Airbus). Some of these facts are curated through manual listing, while others are gathered automatically. In a broader context, the relationship KG denoted as G^R, encompasses pairwise relationships between categories, encompassing "subject-verb-object" connections (e.g., drive and land) and spatial relations (e.g., on and near). The attribute KG, denoted as G^A, is constructed through similarities, organizing aircraft categories based on attributes like color, size, airline, and other characteristics.

The transformation of the concepts and specialized knowledge within the KG into learnable representations is essential, including its relational and attribute-based aspects, in embedded forms. The initial step involves the computation of a $C \times C$ frequency statistics matrix R^c wherein the counts stem from semantic information or co-occurrences among category pairs. Subsequently, to create a class-to-class undirected graph G^R grounded in factual evidence, the transpose $(R^c)^T$ is incorporated into R^c. Lastly, column–row normalization is executed to yield e^R_{ij}, where S_{ii} signifies the summation of the ith row of R^c, and S_{jj} represents the summation of the jth column of R^c

$$G^R : e^R_{ij} = \frac{R^c_{ij}}{\sqrt{S_{ii} S_{jj}}}. \tag{7}$$

Similarly, the initial step involves constructing G^A for the attribute KG using ground truths. This is achieved through the computation of the $C \times A$ frequency statistics matrix, where A signifies attributes and C denotes category targets. After normalization, a symmetric $C \times C$ matrix A^c is created by measuring pairwise Jensen–Shannon divergence between categories, as described by the $C \times A$ frequency distribution table

$$\begin{aligned} G^A : e^A_{ij} &= \mathrm{JS}\left(P_i \| P_j\right) \\ &= \frac{1}{2}\mathrm{KL}\left(P_i \| \frac{P_i + P_j}{2}\right) + \frac{1}{2}\mathrm{KL}\left(P_j \| \frac{P_i + P_j}{2}\right) \end{aligned} \tag{8}$$

where $\mathrm{KL}(X \| Y)$ represents the Kullback–Leibler divergence [41] between the probability distributions X and Y.

2) Feature Enhanced via Knowledge-Based Semantic Reasoning: In ARNet, a total of N_{rp} D-dimensional region proposals have been acquired. The features corresponding to these region proposals can be articulated as $F = \{f_1, f_2, \ldots, f_{N_{\mathrm{rp}}}\} = \{f_i\}_{i=1}^{N_{\mathrm{rp}}}$. Inspired by works on zero/few-shot learning and large-scale object detection [15], [42], [43], which utilize learned models to adapt classifier weights for previously unseen or few categories, the aircraft detection algorithm can employ the classification layer weights as a semantic representation for aircraft categories. Each row of the classifier weights encodes substantial semantic information about instances within a category, thereby contributing to the formation of a global semantic reservoir denoted as $\mathcal{S} \in \mathbb{R}^{C \times D}$, encompassing all C categories. Since the parameters of the classification layer are refined throughout training, the global semantic reservoir $\mathcal{S}$ improves with the progression of training epochs.

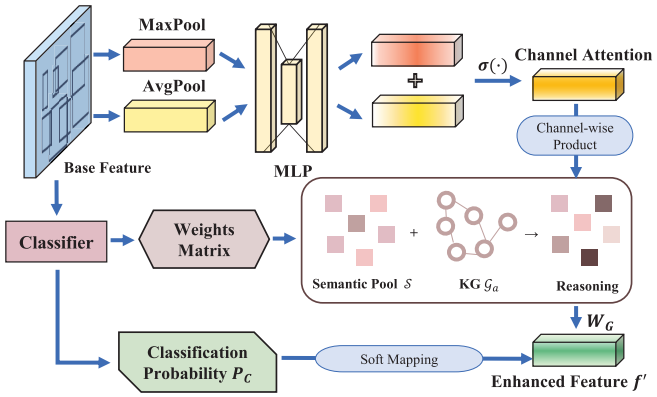


Fig. 4. Illustration of the KRM.

The connections between the semantic reservoir $\mathcal{S}$ and the region proposals are established by an undirected aircraft category-to-category graph, denoted as $\mathcal{G}_a$, is defined as: $\mathcal{G}_a =< \mathcal{N}_a, \mathcal{E}_a >$, which contains category nodes $\mathcal{N}_a$ and relationships $\mathcal{E}_a \in \mathbb{R}^{C \times C}$ between nodes. The prior knowledge embedded in the $\mathcal{G}_a$ is universally propagated through $\mathcal{E}_a$ in the semantic reservoir $\mathcal{S}$ encompassing C categories. This leads to the capture of region features denoted as $\mathcal{E}_a\mathcal{S}$.

The mapping between each region proposal N_{rp} and its corresponding category is typically achieved through direct linkage. However, with the introduction of the KG $\mathcal{G}_a$, additional possibilities for categories corresponding to each region proposal emerge. Consequently, the straightforward direct mapping approach falls short of fully harnessing the potential of prior knowledge within $\mathcal{G}_a$. To effectively leverage this prior knowledge, the mechanism of soft mapping can be employed to associate target classification probabilities with multiple categories to region proposals. The probability distribution $P_c \in \mathbb{R}^{N_{\mathrm{rp}} \times C}$ can be captured by computing the scores of the C classification using the *Softmax* function. This computational process can be expressed as follows:

$$f' = P_c \mathcal{E}_a \mathcal{S} W_G \tag{9}$$

where $W_G \in \mathbb{R}^{D \times K}$ is the transformation weight matrix shared for all graphs and K is the output dimension of the semantic reasoning. This process is shown in Fig. 4.

3) Attention Mechanisms: Knowledge-based global reasoning enhances features while simultaneously introducing a certain degree of category noise. It is noteworthy that not all information is pertinent in the process of aircraft recognition

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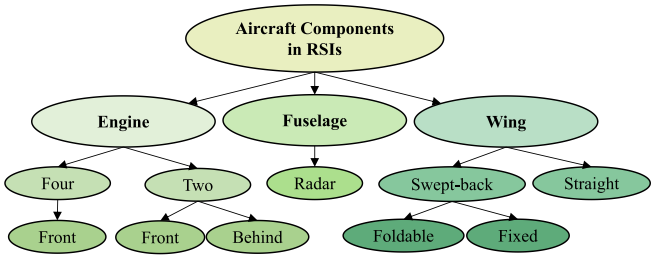


Fig. 5. Illustration of the classification tree of aircraft component features in the RSI.

within a specific image. Experts usually focus on a limited number of potential types during target interpretation. In this study, we employ a convolutional block attention module (CBAM) [44] to dynamically highlight pertinent information and mitigate irrelevant details. More precisely, the entire image's feature map $F_m \in \mathbb{R}^{W\times H\times D}$ serves as input for the computation of both average pooling F_{avg} and maximum pooling $F_{\max}$. In the channel attention process of CBAM, both F_{avg} and $F_{\max}$ are input into the shared multilayer perceptron (MLP) with one hidden layer for generating the attention map $a_c \in \mathbb{R}^{W\times 1}$. To minimize computational parameters, a reduced dimensional coefficient of $r = 64$ is employed in the MLP. The equation for attention is as follows:

$$\begin{aligned} a_c &= \sigma(\text{MLP}(\text{AvgPool}(F_m)) + \text{MLP}(\text{MaxPool}(F_m))) \\ &= \sigma\left(W_{m1}\left(W_{m0}\left(F_{\text{avg}} + F_{\max}\right)\right)\right) \end{aligned} \tag{10}$$

where σ denotes the sigmoid function, $W_{m0} \in \mathbb{R}^{W/r\times W}$, and $W_{m1} \in \mathbb{R}^{W\times W/r}$ denote the weights of MLP. The *Softmax* function is then applied to obtain the category attention: $A_c = softmax(a_c W_s \mathcal{S}^T)$, where $W_s \in \mathbb{R}^{W\times D}$ is weight of the fully connected layer and $A_c \in \mathbb{R}^{C\times 1}$. The enhanced feature f' with adaptive attention reasoning can then be formulated as follows:

$$f' = P_c(A_c \otimes \mathcal{E}_a \mathcal{S}) W_G \tag{11}$$

where $\otimes$ denotes the channel-wise product. The original region feature f in F and the adaptive enhanced feature f' are concatenated, and then the features $f_c = [f; f']$ are transmitted to the bounding box regression layer and classification layer. By sharing edge connections in the KG, f_c acquires pertinent attributes and relations during this process. Furthermore, the addition and discovery of adaptive contexts from the global semantic reservoir under the semantic guidance of prior knowledge.

D. Aircraft Component and Airport Facility

The RSIs are captured from an aerial perspective, which restricts the view of aircraft parked at air bases or commercial airports to their dorsal features and ground projections. These images primarily encompass visual features like the top line of the fuselage, the count of engines, and the configuration of the wings and tail. Nevertheless, specific distinctive features, observable only from frontal or lateral views, such as window shape (pentagon/quadrilateral), nose curvature (streamlined/rounded/concave-convex), and engine

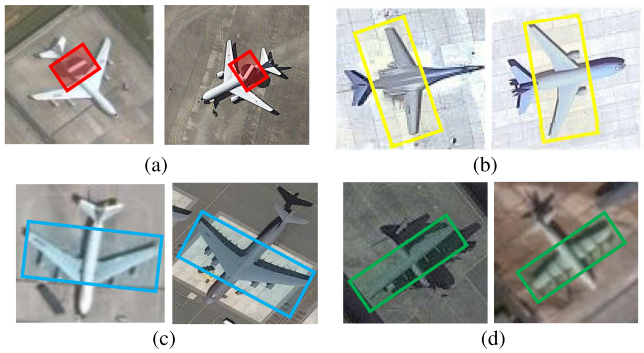


Fig. 6. Illustration of classification tree based on prior knowledge of aircraft features in RSIs. (a) Aircraft with a distinctive fuselage feature. (b) Aircraft with two engine assemblies on the wing, either forward or aft of the wing. (c) Aircraft with four engines mounted in front of or behind the wing. (d) Aircraft with four engine assemblies arranged linearly on the wing.

category (turbine engine/piston engine/ramjet engine, etc.), are not captured within the scope of RSIs. To address the issue of limited detailed features and to incorporate expert prior knowledge, which experts use to identify aircraft types, this article introduces ACDM and AFRM.

1) Aircraft Component Discrimination Module: Following preliminary algorithm feasibility research, as well as observation and analysis, the characteristics of aircraft in RSIs have been categorized into a classification tree, illustrated in Fig. 5. This classification tree segregates features based on varying methods of component location, arrangement, and morphology, as depicted in Fig. 6.

Due to the considerable geometric variations in component scale, posture, and contour, the utilization of deformable convolution (DCN) and DCNv2 [45] are preferred over traditional CNN layers for representing component features. DCN is capable of adapting to the geometric changes of targets and offers a wide range of modeling capabilities. We denote the convolution kernel as s with sampling positions, W_s representing its weights, and B_s representing the preset offset. The dilation of the convolutional kernel is b, which leads to a dilated convolution kernel field size of $(2b-1)^2$

$$B_s(b) = \left\{ \begin{matrix} (-b,b) & (0,b) & (b,b) \\ (-b,0) & (0,0) & (b,0) \\ (-b,-b) & (0,-b) & (b,-b) \end{matrix} \right\}. \tag{12}$$

The input and output features at each location s_k are denoted as $I(s_k)$ and $O(s_k)$, respectively. The offset at each position is represented by b_s. The learnable offset and modulation scalar of position s_k are denoted by Δb_s and Δm_s, respectively. The following mathematical expression represents the modulated DCN:

$$O(s_k) = \sum_{s=1}^{s} W_s \cdot I(s_k + b_s + \Delta b_s) \cdot \Delta m_s \tag{13}$$

where the range of the modulation scalar Δm_s is $[0, 1]$, and b_s is an unbounded real number.

The labeling box for the wing exhibits a significant aspect ratio, potentially resulting in an excessive inclusion of background when employing the HBB labeling method. As a solution, the Faster RCNN-OBB method [46], as proposed in the DOTA dataset, leverages the OBB approach chosen.

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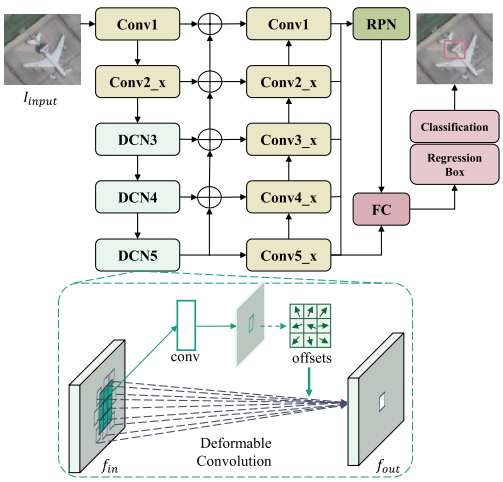


Fig. 7. Illustration of ACDM implementing the Faster RCNN-OBB baseline that includes FPN and DCN.

In the ACDM module, the network structure is built upon a fork of DCNs and uses the Faster-RCNN [47] part of it and makes some modifications based on Faster-RCNN to regress a quadrangle in Fig. 7. The training process involves the utilization of the stochastic gradient descent optimizer. The initial learning rate is set to 0.001 with the warming up for 500 iterations, and the learning rate is decreased by a factor of 0.1 at each decay step. The momentum is set to 0.9, and weight decay is 10^{-4}. The model has been trained for 160 epochs on the AR-CTD dataset.

The results of aircraft components were obtained through the ACDM and established the relationship between the target and its components using a relative position calculation method [23]. First, the prediction score S_{pred} exceeds the predefined threshold λs, which is set at 0.5 based on empirical experience. Second, the predicted components have to belong to the aircraft targets, and the overlap ratio R_{gth} needs to be higher than 0.8. This consideration accounts for potential bias introduced by the aircraft's varying head-to-tail orientation. Afterward, the overlap ratio R_{cst} between the predicted box and a constraint region is calculated. It is essential to note that aircraft may have variable orientations, making it challenging to estimate the relative positions of their components accurately. Therefore, the constraint region is defined with its center aligned to the ground truth box's center. The dimensions of the constraint region are determined by adding ten percent to both the length and width of the ground truth box. S_{ac} is the overall score

$$S_{\text{ac}} = \begin{cases} 0, & S_{\text{pred}} < 0.5 \text{ or } R_{\text{gth}} < 0.8 \text{ or } R_{\text{const}} < 0.5 \\ S_{\text{det}}, & S_{\text{pred}} \geq 0.5 \text{ or } R_{\text{gth}} \geq 0.8 \text{ or } R_{\text{const}} \geq 0.5. \end{cases} \tag{14}$$

2) Airport Facility Relationship Module: The significance of background cannot be overlooked in the intricate cognitive process of human target recognition. Cognitive psychology research [48] reveals that humans, when recognizing targets, engage in a comparison between the target and its background, effectively isolating the focal point from the interplay of the target and its surroundings, thereby enhancing target comprehension. Given the intricate and dynamic landmark scenario, the spatial relationship between airport facilities and aircraft holds paramount significance for the precise recognition of fine-grained aircraft categories.

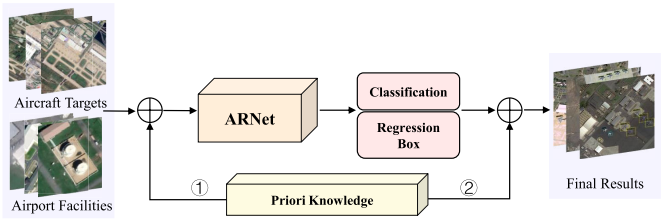


Fig. 8. Illustration of AFRM optimizing the spatial position of aircraft through prior knowledge and network training process.

The majority of algorithms designed for aircraft detection in RSI have traditionally focused solely on aircraft targets, disregarding airport landmarks such as runways, hangars, and fuel storage tanks. Individual airports boast unique landmark structures and facilities that can be logically inferred from the geographic location of the airport in combination with relationships. In the experiments conducted with the AFRM, two methods are employed to enhance aircraft detection results, as illustrated in Fig. 8. The first approach involves integrating prior knowledge extracted from ground facilities into the knowledge reasoning process. The second method entails labeling specific airport facilities and refining the spatial position optimization of aircraft targets during the training process. By incorporating airport facilities into network training and utilizing specialized prior knowledge to fine-tune detection outcomes, it becomes feasible to achieve enhanced detection and recognition in specific environmental conditions.

IV. Experiment

A. Dataset and Setting

1) AR-RSI Dataset: We utilize the self-constructed AR-RSI dataset, which comprises the RS target in this study. The AR-RSI dataset encompasses 143 RSIs obtained from 14 diverse domestic and international panoramic airport locations, with resolutions ranging from 0.5 to 1 m, sourced from the platform *Bigemap* [49]. The distribution of the number of images specifically covered by the AR-RSI dataset in Table I. The AR-RSI annotation format employs a combination of HBB and OBB [8] to accurately label the ground truth. Divided into three distinct sections, the AR-RSI dataset facilitates target detection experiments: the aircraft target detection dataset AR-ATD, the aircraft component detection dataset AR-CTD, and the airport facility detection dataset AR-AFD. The AR-ATD and AR-CTD datasets incorporate both HBB and OBB formats, accommodating the diverse requirements of various DL frameworks. Conversely, the annotation format for AR-AFD only utilizes HBB.

To evaluate the effectiveness and generalization capability of each module in the ARNet algorithm, experiments were conducted on aircraft targets from 24 categories within the AR-ATD dataset of the AR-RSI dataset. In experiments related to the ACDM, ten categories from the AR-CTD dataset were employed. Additionally, experiments involving the AFRM

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TABLE I
DISTRIBUTION OF THE NUMBER OF IMAGES SPECIFICALLY COVERED BY THE AR-RSI DATASET

Raw Data	AR-ATD	AR-CTD	AR-AFD
Panoramic RSI	143	121	32
Training Scene	2921	2650	592

TABLE II
DETAILS OF THE AR-RSI DATASET USED IN THE ABLATION STUDIES

AR-RSI	Category	Train	Validition	Test	Annotation method
AR-ATD	24	1604	101	207	HBB/OBB
AR-CTD	10	831	143	160	OBB
AR-AFD	3	439	60	93	HBB

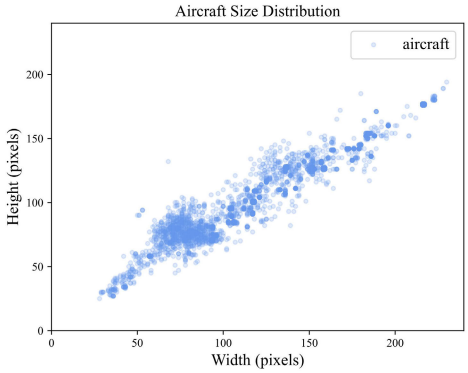


Fig. 9. Illustration of the size distribution of aircraft targets in the AR-RSI dataset. All the targets in the dataset are meticulously labeled and thoroughly inspected through manual efforts. The bounding box aspect ratios exhibit linearity, indicating that fine-grained aircraft targets vary in size while maintaining a consistent ratio of wingspan to fuselage.

utilized data from airport fuel storage tanks and hangars within the AR-AFD dataset. A comprehensive breakdown of the experimental dataset is provided in Table II.

The AR-ATD dataset encompasses 32 distinct categories of aircraft targets, spanning a broad spectrum of aircraft types such as transports, AWACS, refueling planes, bombers, short/medium/long haul airliners, commercial aircraft, and small fighters. This dataset comprises multiple categories of aircraft targets at varying scales. The distribution of scales for the targets is depicted in the scatter plot shown in Fig. 9. Segmented into 32 categories, slices of aircraft targets are allocated based on the training scenes, amounting to a cumulative count of 3846 instances. Considering the range of resolutions for RSIs acquired from *Bigemap*, spanning from 0.5 to 1 m, certain intricacies in the detailed attributes of aircraft may not be clearly distinguishable. Civil passenger planes have been categorized as either long-haul, medium-haul, or short-haul passenger planes to account for this potential ambiguity.

The AR-CTD dataset encompasses annotations for components across 15 distinct categories of aircraft targets. To uphold dataset precision, manual annotation is employed for the AR-RSI dataset. Figs. 5 and 6 illustrate that the majority of category strength features are concentrated around the

TABLE III
EFFECT OF UTILIZING VARIED PRIOR KNOWLEDGE

Prior Knowledge	mAP	mAP_50	mAP_75
Attribute	85.5	95.8	90.1
Relation	85.7	96.1	90.1
All	**86.5**	**98.3**	**91.2**

TABLE IV
ABLATION EXPERIMENTS ON THE VALIDITY OF MODULES IN AN END-TO-END ARNET BASED ON PRIOR KNOWLEDGE-BASED REASONING

Method and Modifications	mAP	mAP_50	mAP_s	mAP_m
No SCM	85.5	96.8	73.7	85.6
No KRM	83.2	94.6	71.7	84.1
No attention mechanism	84.1	95.7	72.5	84.9
The proposed ARNet	**86.5**	**98.3**	**75.2**	**87.2**

wings of aircraft. However, manual labeling of components is resource-intensive, and discerning the core component traits of small aircraft and many passenger planes from RSI presents challenges. It is important to note that not all aircraft within the AR-RSI dataset are covered by this component annotation process.

The AR-AFD dataset currently encompasses several international airports, including a total of 32 panoramic RSI of airport environments. These images are annotated with three distinct types of targets, which include fuel storage tanks and hangars.

2) Implementation Details: In the experiments, four Titan X GPUs were used to train the model. The training image size was set to 1024 × 1024, and data augmentation with horizontal image flipping, cropping, and random contrast was implemented. The same mean global pooling and shared fully connected layers were used in the KRM phase to update the extraction of the region proposals. The training process was optimized using Adam with an initial learning rate of 0.001, an exponential decay rate of 0.9 for the first-moment estimate, and 0.999 for the second-moment estimate. The learning rate was gradually reduced from slow to fast to slow using a cosine annealing strategy [14]

$$\eta_t = \eta^x_{\min} + \frac{1}{2}\left(\eta^x_{\max} - \eta^x_{\min}\right)\left(1 + \cos\left(\frac{T_{\text{cur}}}{T_x}\pi\right)\right) \tag{15}$$

where η_t is the learning rate at step t, $\eta^x_{\min}$ and $\eta^x_{\max}$ are the minimum and maximum values of the learning rate, respectively, T_{cur} is the number of epochs currently executed, and T_x is the total number of epochs in the xth run.

For the aircraft target detection task in ARNet, it is crucial to conduct a comprehensive evaluation of both the regression boxes containing the targets and the detected aircraft categories. The mean average precision (mAP) is the widely employed evaluation metric, which provides a single value to assess its overall performance in object detection tasks. The average precision (AP) is calculated based on both recall

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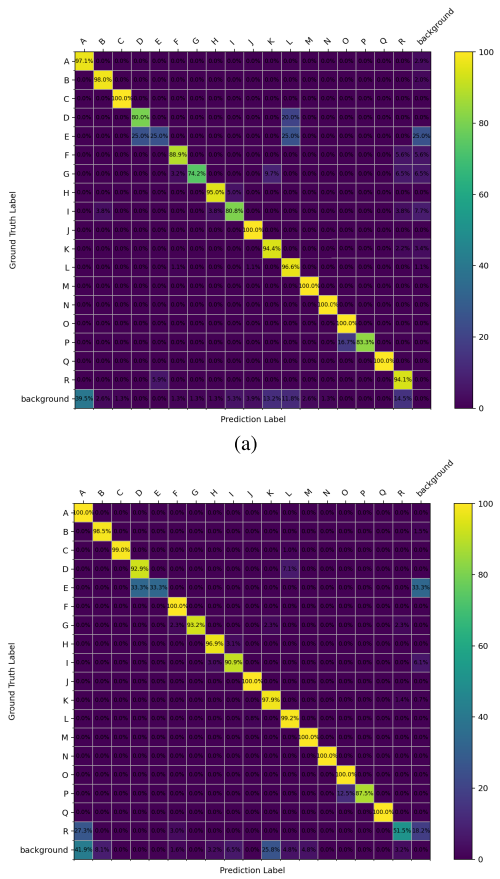


Fig. 10. Confusion matrix illustrates the aircraft target detection and recognition outcomes within the training scenes. (a) Results obtained without prior knowledge. (b) Results achieved with the incorporation of prior knowledge.

TABLE V
EFFECT OF ACDM

Component	B_c	C_c	D_c	H_{c1}	H_{c2}
$AP(\%)$	98.97	95.13	90.78	98.74	64.4
Component	I_c	J_c	K_c	L_c	M_c
$AP(\%)$	94.2	98.19	94.87	93.9	97.81

rate and precision rate. The evaluation equations are provided below

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{16}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{17}$$

$$\mathrm{AP} = \int_0^1 P(R)dR \tag{18}$$

$$\mathrm{mAP} = \frac{\sum \mathrm{AP}_C}{N_C} \tag{19}$$

where TP, FP, FN, P, and R stand for true positive, false positive, false negative, correct, and recall, respectively.

B. Ablation Studies

The effectiveness of the proposed method is discussed in this section, and its performance is evaluated by means of ablation studies for several parts included in the framework and the integration mechanisms used. In addition, the performance of the AFRM and the aircraft component recognition module are evaluated separately.



Fig. 11. Illustration of the detection and recognition results of the ACDM.

TABLE VI
EFFECT OF AFRM

Airport Facility	ARNet without AFRM	ARNet with AFRM
Panoramic Airport 1	78.3%	87.5%
Panoramic Airport 2	94.8%	100%

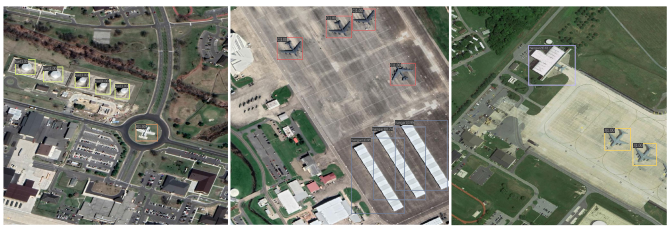


Fig. 12. Illustration of the three types of airport facilities used in the experiment: fuel storage tanks and two variations of hangars.

TABLE VII
COMPARISON OF CATEGORY AND OVERALL PERFORMANCE ON THE AR-RSI DATASET

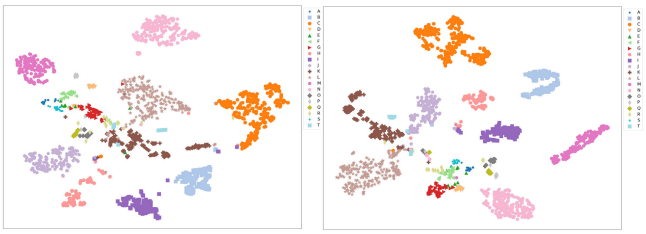
Baseline	Backbone	ACDM	AFRM	mAP%
Faster RCNN	ResNet-101-FPN			82.79
	ResNet-50-FPN			83.55
	ResNet-50-FPN	✓		85.62
YOLOF	ResNet-50-SiMo			85.31
	ResNet-50-SiMo	✓		86.76
ARNet	ResNet-101-FPN			86.52
	ResNet-50-FPN			87.76
	ResNet-50-FPN	✓		90.46
	ResNet-50-FPN	✓	✓	92.91

1) Effect of Utilizing Varied Prior Knowledge: We conducted a thorough analysis of the impact of integrating attribute prior knowledge and relational prior knowledge into ARNet on the overall performance of aircraft target detection. The outcomes of these separate evaluations are outlined in Table III. The experimental findings from the AR-ATD dataset reveal that leveraging attribute prior knowledge enhances the overall detection and recognition performance. Additionally, the utilization of relational prior knowledge exhibits a slight superiority over attribute prior knowledge. Remarkably, the combined integration of both relational and attribute prior

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TABLE VIII
DETECTION AND RECOGNITION ACCURACY OF 24 AIRCRAFT CATEGORIES IN AR-ATD

%	A	B	C	D	E	F	G	H	J	I	K	L
Faster RCNN	45.4	87.7	85.3	90.2	63.7	80.1	81.8	81.4	82.1	71.3	76.5	89.3
YOLOF	60.4	89.2	86.5	92.5	70.4	87.9	82.6	84.1	83.7	**77.8**	82.9	89.9
ARNet	**68.1**	**89.2**	**88.3**	**100.0**	**75.6**	**91.3**	**88.9**	**90.6**	**91.8**	76.7	**86.8**	**95.2**
%	M	N	O	P	Q	R	SHPP	MHPP	LHPP	CA	CA-S	C-other
Faster RCNN	80.7	77.6	92.5	61.7	63.1	74.6	70.9	56.6	34.7	72.2	72.9	47.5
YOLOF	84.2	80.2	94.4	69.7	77.4	**77.8**	72.6	59.1	43.7	77.3	75.9	45.4
ARNet	**93.1**	**82.9**	**96.0**	72.7	**83.4**	72.2	**76.6**	**59.5**	**46.7**	**78.7**	**79.2**	**50.8**



(a) (b)

Fig. 13. 2-D visualization of features using the t-SNE algorithm: (a) features without prior knowledge and (b) features with prior knowledge.

knowledge leads to a substantial enhancement in overall performance.

2) Effect of SCM and KRM in the End-to-End Network: The end-to-end network architecture in our proposed framework comprises two primary modules. The first module optimizes the localization of potential region proposals for the target through SCM. Subsequently, global semantic reasoning is conducted using KRM. Notably, KRM incorporates soft mapping and channel attention mechanisms to enhance region proposal features. As illustrated in Table IV, SCM can improve mAP to 1.4%–2%, KRM can significantly improve mAP to 3.1%–3.7%, and the attention mechanism in KRM can improve mAP by 2.3%–2.7%. The experimental results on the AR-RSI dataset show that removing either SCM or KRM decreases the overall results compared to the network with the two modules. Moreover, the soft mapping and channel attention mechanisms facilitate the global semantic reasoning of the KRM effectiveness.

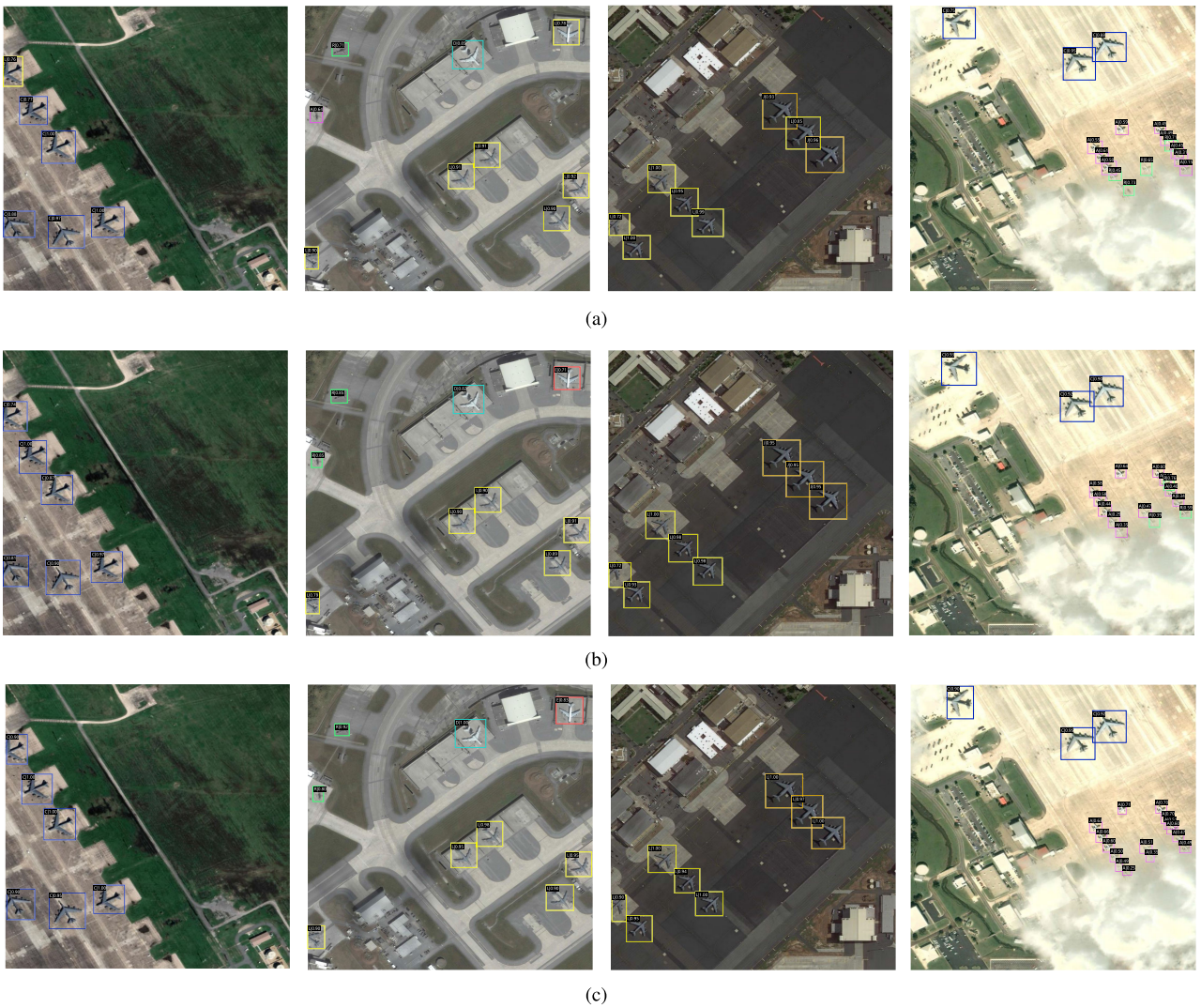
Furthermore, the confusion matrix in Fig. 10 illustrates the notable influence of employing a knowledge-based reasoning module on the outcomes of detection and recognition. Observing confusion matrix (a), it becomes evident that before the inclusion of prior knowledge, confusion is prevalent among numerous aircraft categories. However, in (b), the state of confusion is notably diminished or entirely eliminated. Category R, while categorized as a fine-grained aircraft target, actually encompasses a broader concept during the practical experimental phase, referring to the limited count of unlabeled aircraft at airports. Consequently, within the KG, there is no node capable of representing these hybrid aircraft entities. Furthermore, category R is suppressed and weakened after the self-channeling attention enhancement of the attention mechanism, ultimately resulting in reduced accuracy.

3) Effect of ACDM: Due to the diverse characteristics of aircraft targets, incorporating component discrimination is considered to enhance the network's ability to recognize aircraft targets. The ARNet framework not only integrates end-to-end aircraft target detection algorithms, but also introduces additional modules to create more accurate target models. As detailed in Table V, experiments are conducted on the AR-CTD subset of the AR-RSI dataset to detect and recognize various categories of aircraft components. Fig. 11 shows the effect of ACDM detection and recognition. Most of the components examined were in the wings and fuselage, with only the H_{c2} head position. However, it was not utilized due to its limited effectiveness. The raw results are then combined with the component-based outcomes using the ACDM formulation outlined in (14). Drawing from the expert interpretation's prior knowledge, focusing on specific components in Fig. 6 becomes a viable strategy.

4) Effect of AFRM: In AFRM, the ARNet framework leverages prior knowledge related to airport geographical information, specifically addressing fuel storage tanks and two types of hangars, and their associations with aircraft. As detailed in Table VI, the experiments were performed on the AR-AFD subset of the AR-RSI dataset. As shown in Fig. 12, the outcomes of these experiments highlight the efficacy of AFRM.

The results of employing the AFRM are summarized in Table VI. The AR-AFD dataset refers to the labeled dataset containing information about fuel storage tanks and hangars within the two airports presented in Table VI. During the learning process, the relationships between fuel storage tanks or hangars and the airplanes located in their proximity are strengthened through the edge links within the KG. Simultaneously, the soft-mapping mechanism within the KRM ensures that even low-probability and infrequently occurring airplane models are retained. Therefore, this reasoning process significantly enhances the accuracy of classifying targets near the labeled airport facilities, regardless of whether they are strongly associated targets or occasional occurrences. Furthermore, AFRM can be applied as a postprocessing step for airports with simpler distributions to further improve target recognition accuracy. The geographic information within the

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(a)

(b)

(c)

Fig. 14. Illustration of results for fine-grained aircraft target detection and recognition. (a) Outcome of Faster RCNN. (b) Results of YoloF. (c) Results obtained using the ARNet method introduced in this article. ARNet demonstrates strong performance in aircraft localization, classification, and target confidence.

KG, combined with expert inspections, serves to rectify the subsequent detection results. For instance, it ensures that aircraft from Country Y will not be erroneously categorized as belonging to Country Z, thereby resolving any confusion regarding aircraft types. Similarly, for bases such as Base M, which are primarily associated with logistics and transportation, the presence of bombers is highly unlikely. This correction process aligns the detection results with the expected categorizations based on geographic and contextual relationships.

C. Comparison With State-of-the-Arts

We present a comparison of experimental results using 24 aircraft target categories from the AR-RSI dataset, in Table VII. We compare the outcomes of ARNet, the method proposed in this article, with those of other target detection networks, including Faster RCNN with FPN [50] and YoloF [51] target detection networks. All experiments were conducted with the same hyperparameter settings unless explicitly stated in the implementation details. Given the large dataset, two backbone networks, ResNet50 and ResNet101, were employed to achieve improved experimental outcomes. During the experiment, the target training data with ground-truth labels encompass a total of 1912 airport slices, encompassing 14 domestic and international airports as detailed in Table II. The dataset is partitioned into 1604 training image slices, 101 validation set image slices, and 208 test image slices based on airport assignment.

It is obvious from Table VII that ARNet boosted the accuracy of aircraft target detection and recognition compared to other algorithms, achieving an overall mAP of 92.91%. ARNet with prior knowledge outperforms the baseline Faster RCNN with FPN on the AR-RSI dataset, with 4.21% performance improvement when using ResNet50 as the backbone and 3.73% when using ResNet101 as the backbone. ResNet50 achieved comparatively better results, so it was used for all ARNets where the backbone was not mentioned. Furthermore, by ACDM and AFRM, a significant performance increase of 9.36% can be achieved. The experiments also included a comparison of ARNet with the mainstream single-stage

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YPLOF target detection network for aircraft target detection. The experimental result of ARNet compared to YOLOF is improved by 3.7% for both using the ACDM, and the complete ARNet algorithm can improve the results by 7.6% compared to YOLOF.

As depicted in Table VIII, ARNet significantly improves detection and fine-grained recognition accuracy compared to alternative algorithms. The experimentation covered 24 categories of aircraft targets within AR-RSI, including 18 categories at the fine-grained level and six categories at the coarse-grain level. Within the 18 fine-grained target categories depicted in the confusion matrix, the lower AP scores in the I and P categories stem from their close association with the Boeing series, confusing with the coarse-grained SHPP aircraft. As demonstrated in the confusion matrix in Fig. 10, category R is because there are no knowledge nodes in the fine-grained aircraft KG that correspond to the hybrid category of aircraft. It is noteworthy that category A targets represent small fighter aircraft with real-world dimensions falling within the range of 15–20 m. In the AR-RSI dataset, there were originally three subcategories within the small fighter category. However, due to limited data samples, these three subcategories were merged into one for the experimental analysis. Clustering of visualized target features with t-SNE [52] algorithm in Fig. 13. Fig. 14 displays the results of fine-grained aircraft target detection and fine-grained recognition. This vividly illustrates ARNet's superior performance in fine-grained aircraft recognition, particularly when dealing with long-tailed data, intricate backgrounds, and varying alignment densities.

To gain a deeper understanding and visualize the enhanced representation of target features constructed by ARNet using prior knowledge, the output feature vectors of the proposed method are captured during the recognition of 18 fine-grained target categories, both with and without the incorporation of prior knowledge. The features exhibit a more relaxed clustering pattern when compared to the configuration where all prior knowledge in ARNet remains empty, as depicted in Fig. 13(a). The results (b) from t-SNE illustrate that the introduction of prior knowledge into ARNet leads to a reduction in intra-class distance and a more pronounced aggregation of similar features.

V. Conclusion

This article introduces ARNet, an ARNet based on prior knowledge. ARNet encompasses four pivotal modules: the KRM designed to establish semantic relational connections among small and frequent samples, enabling an adaptive realization of the target knowledge reasoning process, the SCM dedicated to preserving high-confidence aircraft locations to accommodate varying target densities, the ACDM tailored to enhance classification performance for few-shot categories and challenging aircraft categories, and the AFRM devised to uncover spatial relationships between aircraft targets and fixed airport facilities, thereby adapting to diverse environmental backgrounds. Notably, KRM, SCM, and AFRM can be seamlessly integrated into the end-to-end detection network framework, while ACDM functions as an independent module. The prior knowledge of aircraft targets, the tree diagram depicting aircraft components, and the specific structure outlined in this article have been meticulously curated through extensive experience. It is crucial to note that these elements are not haphazardly appended pieces of knowledge or components. Forthcoming work will consider the integration of these components into oriented box target detection algorithms. Furthermore, the application will be extended to include not only RSIs, but also synthetic aperture radar (SAR) image target detection and recognition algorithms.

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